

Anjani Kumar Maurya

Associate Staff Scientist

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Research and Work Experience

SLAC National Accelerator Laboratory, Stanford University, California, USA

Associate Staff Scientist

Dec 2025 – Present

Manager: Prof. Christopher J. Tassone

- Developing a soft matter research group that utilizes large-scale facilities such as synchrotrons and free-electron lasers (FELs).
- Developing autonomous experimental platforms integrating X-rays capabilities at the SSRL beamline using AI/ML.

SLAC National Accelerator Laboratory, Stanford University, California, USA

Postdoctoral Research Fellow

May 2021 – Nov 2025

Supervisor: Prof. Christopher J. Tassone

- Performed multiscale structural characterization of bioplastics and recyclable polymers to elucidate nanoscale–macroscale structure–property relationships.
- Developed experimental platforms at SSRL (1-5, 17-2, 11-3) integrating SAXS/WAXS with in-situ tensile, heating, microfluidics, and reaction environments.
- Built Python-based real-time data reduction and analysis pipelines for scattering experiments, enabling closed-loop and active learning workflows.
- Conducted ultrafast pump–probe X-ray scattering at LCLS (XPP) to probe transient melting and structural dynamics in metal films.

Empa, Swiss Federal Laboratories for Materials Science and Technology (ETH Domain), Switzerland

PhD Thesis Research

Mar 2017 – Mar 2021

Supervisors: Prof. Alex Dommann & Prof. Martin Frenz

- Advanced expertise in SAXS/WAXS at synchrotrons and lab instruments, designing in-situ experiments with microfluidic, tensile, thermal, electrical, and humidity stimuli.
- Investigated nanoparticle nucleation and growth via microfluidic synthesis using X-ray scattering.
- Built quantitative structure–property relationships in fibers and nanofibrous polymeric materials with focus on orientation–mechanics links.
- Performed large-scale data analysis and modeling of hierarchical soft matter structures.

Infineon Technologies AG, Neubiberg, Germany

Materials Simulation Engineer (Part-time), Package Simulation Team

Nov 2015 – Feb 2017

Supervisor: Dr. Martin Niessner

- Developed MATLAB/Python tools to analyze experimental data and benchmark simulations.
- Built material models using FEM tools (COMSOL Multiphysics, Ansys).
- Maintained a materials database to support packaging simulations.

Technical University of Munich (TUM), Germany

Master Thesis Project

Oct 2015 – Nov 2016

Supervisor: Prof. Peter Müller-Buschbaum

Thesis: Optimization of the thermoelectric properties of conducting polymer thin films by dedoping and hybrid approaches

- Fabricated conducting polymer thin films and enhanced thermoelectric performance through structure–property analysis.

Laboratoire Kastler Brossel-UPMC-ENS, Paris, France

Summer Internship

May 2015 – Aug 2015

Supervisors: Prof. Quentin Glorieux & Prof. Alberto Bramati

Internship: Hybrid apparatus: Optical microfiber knot/loop resonator

- Fabricated microfiber knot/loop resonators for quantum dot coupling to study light–matter interactions.

Indian Institute of Technology Guwahati (IITG), India

B.Tech Thesis Project, Semiconductor Group

Jan 2014 – May 2014

Supervisor: Prof. Pratima Agarwal

Thesis: Fabrication and characterization of silicon ultra-thin film prepared by CVD techniques

- Fabricated ultra-thin silicon by HWCVD and PECVD; performed electrical, optical, and structural characterization.

IIT Guwahati, India

Research Project, Semiconductor Group

May 2013 – Dec 2013

Supervisor: Prof. Pratima Agarwal

Project: Band gap variation in bilayer graphene and silicene under transverse electric field

- Used DFT to demonstrate higher band gap tunability of silicene over graphene for potential room-temperature FETs.

IIT Guwahati, India

Research Project, Magnetism Group

Jan 2013 – May 2013

Supervisor: Prof. D. Pamu

Project: Fabrication of $K_{0.05}Na_{0.05}NbO_3$ (KNN) ferroelectric thin film capacitor by RF magnetron sputtering

- Synthesized KNN and deposited dual capacitor layers by RF sputtering for Pb-free ferroelectric/piezoelectric devices.

IIT Guwahati, India*Summer Internship, Semiconductor Group*

May 2011 – Jul 2011

Supervisor: Prof. Pratima Agarwal

Project: Structural and transport studies of ZnS thin film for photovoltaic applications

- Synthesized ZnS nanoparticles via solvothermal processing and fabricated optimized thin films for PV applications.

Education

Stanford University Graduate School of Business, Stanford, USA

Jun 2023 – Jul 2023

Stanford Ignite – Full-Time

- Exposure to marketing, operations, strategy, accounting, finance, economics; practiced design thinking, teamwork, leadership, and pitching; developed a commercialization plan for a new venture.

PhD, Empa (ETH Domain) & University of Bern, Switzerland

Mar 2017 – Mar 2021

Biomedical Engineering, *Summa Cum Laude*

Thesis: Multiscale structural decoding of fibers and designing responsive nanofibrous materials

Supervisors: Prof. Alex Dommann & Prof. Martin Frenz; Co-advisor: Prof. Antonia Neels

MSc (M2), Technical University of Munich (TUM), Germany

Oct 2015 – Nov 2016

Applied and Engineering Physics, Erasmus Mundus (MaMaSELF)

Thesis: Optimization of thermoelectric properties of conducting polymer thin films by dedoping and hybrid approaches

Supervisor: Prof. Peter Müller-Buschbaum

MSc (M1), University of Rennes 1, France

Sep 2014 – Aug 2015

Physics, Erasmus Mundus (MaMaSELF)

Mentor: Prof. Sergio Di Matteo

B.Tech, Indian Institute of Technology Guwahati (IITG), India

Jul 2010 – Jun 2014

Engineering Physics

Thesis: Fabrication and characterization of silicon ultra-thin film prepared by HWCVD

Supervisor: Prof. Pratima Agarwal

Skills

Core: Materials Synthesis (electrospinning, melt-spinning, microfluidics, spin coating, 3D printing, thin film deposition: CVD, evaporation); X-ray (SAXS, WAXS, ultrafast pump-probe diffraction, XAS); Programming & Analysis (Python, MATLAB, real-time SAXS/WAXS pipelines); Instrumentation (beamline automation, in-situ setups); Modeling & Simulation (FEM, DFT).

Software: TOPAS, ATSAS, NIKA, FIT2D, GIXGUI, DPDAK, SasView, SASfit, BioXTAS RAW, PyMOL, Fusion 360.

Instrumentation Design & Beamline Automation

- Experimental work at DESY (P12), PSI (MS), DLS (I15-1), SSRL (1-5, 17-2, 11-3, 4-2, 9-3, 4-1), and LCLS (XPP).
- Python pipelines to automate SAXS/WAXS reduction, correction, and real-time analysis.
- Integrated Anton Paar TS600 tensile setup with SSRL 17-2 for efficient synchronized acquisition.

- High-temperature, high-pressure capillary setup at SSRL 1-5; multiple in-situ setups for lab SAXS/WAXS; training and maintenance for Molmet/Nanostar.

Awards and Honors

- Young Scientist Award to attend SAS2022 International Conference (July 2022).
- IUCr Young Scientist Award (May 2020).
- Best Poster, EXCITE Biomedical Imaging Summer School, ETH Zurich (Sept 2018).
- Erasmus+ Scholarship, European Union (2015–2016).
- World Quantitative and Science Scholarship, World Quant Foundation (2014–2015).
- IIT Guwahati Institute Merit-cum-Means Scholarship (2012–2014).
- Top 0.1% in IIT-JEE 2010 (1 million candidates).

Professional Service

- Peer Reviewer (2024–Present): Journal of Molecular Structure (Elsevier), Micro and Nanostructures (Elsevier), Journal of Cluster Science (Springer Nature), Cellulose (Springer Nature), Scientific Reports (Springer Nature), Soft Matter (The Royal Society of Chemistry (RSC)).
- Session Chair: Colloids and Polymers 3, SAS 2022, Campinas, Brazil.

Teaching and Supervision

- Taught “Applied Biomaterials” to Master’s students (University of Bern), with Prof. Alex Dommann.
- Supervised Master’s thesis of Eloise Mias (University of Rennes 1, MaMaSELF).
- Assisted Prof. Alex Dommann with exams and coursework (University of Bern).

Organizational & Managerial

- Managed and collaborated on interdisciplinary projects; volunteered at Bio-X (2018) and SolTech (2016); TechnoThlon city representative (2011); ICANN volunteer (2011).

Languages

Hindi (native), English (native), German (A2), French (A2).

Memberships

American Physical Society (2024–present); American Chemical Society (2024–present); Swiss Crystallography Society (2019–present); Swiss Chemical Society (2017–2022); German Physical Society (2015–2018).

Publications

See [full list in the Appendix](#) or Google Scholar/ResearchGate.

Conference Activity

Recent talks and posters include APS/ACS 2025 plastics deconstruction via in-situ SAXS/WAXS; SAS2022 autoxidation deconstruction mechanisms; EPDIC17 fibrous materials; multiple Swiss/EU venues. See [Appendix — Conference Proceedings](#) and [Appendix — Conference Presentations](#) for complete lists.

References

Prof. Christopher J. Tassone

Associate Lab Director, Energy Sciences Directorate (ESD), SLAC; formerly Materials Science Division Director, SSRL

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Prof. em. Dr. Alex Dommann

Professor Emeritus, University of Bern; former Head, Materials meet Life, Empa (2013–2023)

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Prof. Dr. Antonia Neels

Head, Center for X-ray Analytics, Empa

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Publications

26. **Anjani K. Maurya**, Ravikumar Ramegowda, Arun S. Asundi, Ozge D. Bozkurt, Katrina Knauer, Christopher J. Tassone; “Decoding the structural deformation behavior of redesigned bioplastics” (*in-preparation*).
25. Arun S. Asundi, **Anjani K. Maurya**, Ozge D. Bozkurt, Simon R. Bare, Ritimukta Sarangi, Christopher J. Tassone; “Understanding effect of water in Co/Mn-catalyzed oxidation of polyethylene using X-ray scattering and spectroscopy techniques” (*submitted*).
24. Taekeun Yoon, Haoyuan Li, Yanwen Sun, Arijit Majumdar, Priyanka Muhunthan, Guillaume Vignat, Sanghoon Song, Jingcun Fan, Selene She, Takahiro Sato, Aliaksei Halavanau, Alberto Lutman, Yuxuan Li, **Anjani K. Maurya**, Diling Zhu, Matthias Ihme; “Revealing Ultrafast Multiscale Dynamics of Liquid CO₂ Radiolysis with Time-Resolved X-ray Pump/X-ray Probe Scattering using Split-and-Delay Optics” *submitted* (2026).
23. **Anjani K. Maurya**, Arun S. Asundi, Sarah A. Hesse, Amani M. Ebrahim, Kevin P. Sullivan, Joel Miscall, Jocelyn A. Richardson, Simon R. Bare, Ritimukta Sarangi, Gregg T. Beckham, Christopher J. Tassone; “Effects of Polymer Morphology on Solvent and Catalyst Accessibility during Polyethylene and Polystyrene Autoxidation” *JACS Au* **6** (2026) 2891–2901, DOI:[10.1021/jacsau.6c00230](https://doi.org/10.1021/jacsau.6c00230).
22. **Anjani K. Maurya**, Arun S. Asundi, Ozge D. Bozkurt, Katrina M. Knauer, Simon R. Bare, Christopher J. Tassone; “Polymer Deconstruction and Redesign Strategies for Plastics Recycling” *Advanced Materials Technologies* **11** (2026) 2502008, DOI:[10.1002/admt.202502008](https://doi.org/10.1002/admt.202502008).
21. Julia B. Curley, Yuanzhe Liang, Jason S. DesVeaux, Taylor Uekert, Hoon Choi, Ryan W. Clarke, William E. Michener, **Anjani K. Maurya**, Lisa M. Stanley, Yue Wu, Sarah A. Hesse, Andrea L. Baer, Hudson A. Neyer, Christopher J. Tassone, Alan J. Jacobsen, Ofel D. Mante, Gregg T. Beckham, Katrina M. Knauer; “Closed-loop recycling of mixed polyesters via catalytic methanolysis and monomer separations” *Nature Chemical Engineering* **2** (2025) 568–580, DOI:[10.1038/s44286-025-00275-x](https://doi.org/10.1038/s44286-025-00275-x).
20. Jessica Lusty Beech, **Anjani K. Maurya**, Ronivaldo da Silva, Emmanuel Akpoto, Arun Asundi, Julia Ann Fecko, Neela Yennawar, Ritimukta Sarangi, Thomas Weiss, Christopher J. Tassone, Jennifer L. DuBois; “Understanding the stability constraints on a plastic-deconstructing Rieske iron oxidoreductase system” *Protein Science* (2024) DOI:[10.1002/pro.4997](https://doi.org/10.1002/pro.4997).
19. Amy A. Cuthbertson, Clarissa L. Lincoln, Joel Miscall, Lisa M. Stanley, David Moore, **Anjani K. Maurya**, Arun S. Asundi, Christopher J. Tassone, Nicholas A. Rorrer, Gregg T. Beckham; “Characterization of polymer properties and identification of additives in commercially available research plastics” *Green Chemistry* (2024) DOI:[10.1039/D4GC00659C](https://doi.org/10.1039/D4GC00659C).
18. Arun S. Asundi, Amani M. Ebrahim, **Anjani K. Maurya**, Chad T. Palumbo, Kevin P. Sullivan, Gregg T. Beckham, Ritimukta Sarangi; “Effect of Ligand Chemistry on Electronic Properties and Reactivity of Cobalt Acetate Autoxidation Catalysts” *The Journal of Physical Chemistry C*, 127 (32), 15797–15808, (2023) DOI:[10.1021/acs.jpcc.3c03216](https://doi.org/10.1021/acs.jpcc.3c03216).
17. Makenna L. Pennel, **Anjani K. Maurya**, Amani M. Ebrahim, Christopher J. Tassone, Matteo Cargnello; “Intrinsic activity of silica-alumina for the conversion of polyethylene into tunable aromatics below pyrolytic temperatures,” *ACS Sustainable Chemistry & Engineering*, 11 (34), 12623–12630, (2023) DOI:[10.1021/acssuschemeng.3c02295](https://doi.org/10.1021/acssuschemeng.3c02295).
16. Fernando Vazquez Luna, **Anjani K. Maurya**, Juliana Martins de Souza e Silva, Guido Dittrich,

- Theresa Paul, Dirk Enke, Patrick Huber, Ralf Wehrspohn, and Martin Steinhart; “Straight versus Spongy: Effect of Tortuosity on Polymer Imbibition into Nanoporous Matrices Assessed by Segmentation-Free Analysis of 3D Sample Reconstructions” *The Journal of Physical Chemistry C*, 126 (30), 12765–12779, (2022) DOI:[10.1021/acs.jpcc.2c01991](https://doi.org/10.1021/acs.jpcc.2c01991).
15. Deeptanshu Sivaraman, Gilberto Siqueira, **Anjani K. Maurya**, Shanyu Zhao, Matthias M. Koebel, Gustav Nyström, Marco Lattuada, Wim J. Malfait; “Superinsulating nanocellulose aerogels: Effect of density and nanofiber alignment” *Carbohydrate Polymers*, 292, 119675 (2022) DOI:[10.1016/j.carbpol.2022.119675](https://doi.org/10.1016/j.carbpol.2022.119675).
 14. J. Schoeller, J. T. Avaro, **Anjani K. Maurya**, R. M. Rossi, A. Neels; “Tailoring Fibre Structure Enabled by X-ray Analytics for Targeted Biomedical Applications” *Chimia* 2022, 76, 229 (2022) DOI:[10.2533/chimia.2022.229](https://doi.org/10.2533/chimia.2022.229).
 13. Robin M. Cywar, Nicholas A. Rorrer, Heather B. Mayes, **Anjani K. Maurya**, Christopher J. Tassone, Gregg T. Beckham, and Eugene Y.-X. Chen; “Redesigned Hybrid Nylons with Optical Clarity and Chemical Recyclability” *Journal of the American Chemical Society*, 144(12), 5366–5376 (2022) DOI:[10.1021/jacs.1c12611](https://doi.org/10.1021/jacs.1c12611).
 12. Iranpour, N., Liebi, M., Ong, Q., Blanchet, C., **Anjani K. Maurya**, Stellacci, F., Salentinig, S., Wick, P., Neels, A.; “In-situ Investigations on Gold Nanoparticles Stabilization Mechanisms in Biological Environments Containing HSA” *Advanced Functional Materials*, 32, 2110253 (2022) DOI:[10.1002/adfm.202110253](https://doi.org/10.1002/adfm.202110253).
 11. **Anjani K. Maurya**, Eloise Miase, Jean Scholler, Giuseppino Fortunato, René M. Rossi, Alex Dommann, Antonia Neels*; “Multiscale structural decoding of electrospun fibers: from fabrication to possibilities for steering properties” *Nanoscale Advances*, 4, 491–501 (2022) DOI:[10.1039/D1NA00503K](https://doi.org/10.1039/D1NA00503K).
 10. Inès Richard, **Anjani K. Maurya**, Shahrzad Shadman, Eloïse Masquelier, Lison Sylou Marthey, Antonia Neels, Fabien Sorin; “Unraveling the influence of polymer chain orientation on the thermo-mechanical properties of thermally drawn fibers” *Small*, 18, 2101392 (2022) DOI:[10.1002/sml.202101392](https://doi.org/10.1002/sml.202101392).
 9. **Anjani K. Maurya**, Annapaola Parrilli, Tatiana Kochetkova, Jakob Schwiedrzik, Alex Dommann, Antonia Neels*; “Multiscale and multimodal X-ray analysis: Quantifying phase orientation and morphology of mineralized turkey leg tendons” *Acta Biomaterialia*, 129, 169–177 (2021) DOI:[10.1016/j.actbio.2021.05](https://doi.org/10.1016/j.actbio.2021.05).
 8. **Anjani K. Maurya**, Sumit Mondal, Dean E. Wheeldon, Jean Scholler, Michel Schmid, Simon Annaheim, Martin Camenzind, Giuseppino Fortunato, Alex Dommann, Antonia Neels, Amin Sadeghpour, René M. Rossi; “Effect of radiant heat exposure on structure and mechanical properties of thermal protective fabrics” *Polymer*, 222, 123634 (2021) DOI:[10.1016/j.polymer.2021.123634](https://doi.org/10.1016/j.polymer.2021.123634).
 7. Dambarudhar Parida, Khalifah Salmeia, Amin Sadeghpour, Shanyu Zhao, **Anjani K. Maurya**, Eva Moreau, Robin Pauer, Sandro Lehner, Milijana Jovic, Sabyasachi Gaan; “Template free synthesis of hybrid mesoporous silica nanoparticle with phosphonic acid functionality for efficient methylene blue removal” *Materials & Design*, 201, 109494 (2021) DOI:[10.1016/j.matdes.2021.109494](https://doi.org/10.1016/j.matdes.2021.109494).
 6. Ivana Malagurski, Ruggero Frison, **Anjani K. Maurya**, Antonia Neels, Boban Andjelkovic, Ram-sankar Senthamarai-kannan, Ramesh Babu Padamati, Kevin O’Connor, Tomasz Witko, Daria Solarz, Jasmina Nikodinovic-Runic*; “Polyhydroxyoctanoate Films Reinforced with Titanium Dioxide Microfibers for Biomedical Application” *Materials Letters*, 285, 129100 (2020) DOI:[10.1016/j.matlet.2020.129100](https://doi.org/10.1016/j.matlet.2020.129100).
 5. Tatiana Kochetkova*, Cinzia Peruzzi, Oliver Braun, Jan Overbeck, **Anjani K. Maurya**, Antonia Neels, Michel Calame, Johann Michler, Philippe Zysset, Jakob Schwiedrzik*; “Combining polarized Raman spectroscopy and micropillar compression to study microscale structure-property relationships

- in mineralized tissues” *Acta Biomaterialia* (2020) DOI:[10.1016/j.actbio.2020.10.034](https://doi.org/10.1016/j.actbio.2020.10.034).
4. Tien N. D., **Anjani K. Maurya**, G. Fortunato, M. Rottmar, R. Zboray, R. Erni, A. Dommann, R. M. Rossi, A. Neels, A. Sadeghpour*; “Responsive Nanofibers with Embedded Hierarchical Lipid Self-Assemblies” *Langmuir*, 36(40), 11787–11797 (2020) DOI:[10.1021/acs.langmuir.0c01487](https://doi.org/10.1021/acs.langmuir.0c01487).
 3. Arushi Jain, **Anjani K. Maurya**, Leonie Ulrich, Michael Jaeger, René M. Rossi, Antonia Neels, Philippe Schucht, Alex Dommann, Martin Frenz, and H. Günhan Akarçay; “Polarimetric imaging in backscattering for the structural characterization of strongly scattering birefringent fibrous media” *Optics Express* 28, 16673–16695 (2020) DOI:[10.1364/OE.390303](https://doi.org/10.1364/OE.390303).
 2. **Anjani K. Maurya**, Lukas Weidenbacher, Fabrizio Spano, Giuseppino Fortunato, René M. Rossi, Martin Frenz, Alex Dommann, Antonia Neels*, Amin Sadeghpour*; “Structural insights into semicrystalline states of electrospun nanofibers: a multiscale analytical approach” *Nanoscale* 11, 7176–7187 (2019) DOI:[10.1039/C9NR00446G](https://doi.org/10.1039/C9NR00446G).
 1. Nitin Saxena, Josef Keilhofer, **Anjani K. Maurya**, Giuseppino Fortunato, Jan Overbeck, Peter Müller-Buschbaum*; “Facile Optimization of Thermoelectric Properties in PEDOT:PSS Thin Films through Acid–Base and Redox Dedoping Using Readily Available Salts” *ACS Applied Energy Materials* 1(2), 336–342 (2018) DOI:[10.1021/acsaem.7b00334](https://doi.org/10.1021/acsaem.7b00334).

Conference Proceedings

4. **Anjani K. Maurya**, Dommann, A. Neels; “Multiscale structural decoding of fibrous materials by SAXS and WAXD”, *Acta Cryst.* (2021). A77, C571 DOI: [10.1107/S0108767321091236](https://doi.org/10.1107/S0108767321091236)
3. **Anjani K. Maurya**, L. Weidenbacher, F. Spano, G. Fortunato, R. M. Rossi, M. Frenz, A. Dommann, A. Sadeghpour and A. Neels; “Multiscale structural decoding of electrospun nanofibres: from processing to possibilities for steering functionality”, *Acta Cryst.* (2019). A75, e647 DOI: [10.1107/S2053273319089095](https://doi.org/10.1107/S2053273319089095)
2. Amin Sadeghpour, Tien N. D., **Anjani K. Maurya**, G. Fortunato, A. Dommann, R. M. Rossi and A. Neels; “Hierarchical design of lipid–polymer composite nanofibres: the interplay of multiscale structures and biofunctions”. *Acta Cryst.* (2019). A75, e587. DOI: [10.1107/S2053273319089691](https://doi.org/10.1107/S2053273319089691)
1. Malagurski, R. Frison, **Anjani K. Maurya**, J. Nikodinovic-Runic, R. Babu, K.E. O’Connor, A. Neels; “Medium chain length (mcl)-PHA-based nanocomposites for biomedical applications: system evaluation through XRD”, *Acta Cryst.* (2019). A75, e577. DOI: [10.1107/S2053273319089794](https://doi.org/10.1107/S2053273319089794)

Conference Presentations

26. **Anjani K. Maurya**, Arun S. Asundi, Ozge D. Bozkurt, Julia Curley, Katrina M. Knauer, Gregg T. Beckham, Christopher J. Tassone; “Decoding Structure–Property Relationships in Redesigned Polymers Using *In Situ* WAXS and Tensile Measurements”; March 2026, APS Global Physics Summit, Denver, USA.
25. **Anjani K. Maurya**, Arun S. Asundi, Ozge D. Bozkurt, Julia Curley, Katrina M. Knauer, Gregg T. Beckham, Christopher J. Tassone; “Chemical Deconstruction of Plastics: Insights from *in situ* simultaneous SAXS and WAXS”; March 2025, APS Spring Meeting, Anaheim, USA.
24. **Anjani K. Maurya**, Arun S. Asundi, Sarah A. Hesse, Amani M. Ebrahim, Kevin P. Sullivan, Joel Miscall, Simon R. Bare, Ritimukta Sarangi, Gregg T. Beckham, Christopher J. Tassone; “Unraveling the impact of solvents and catalysts on plastics in catalytic autoxidation: Insights from X-ray scattering”; March 2025, ACS Spring Meeting, San Diego, USA.

23. **Anjani K. Maurya**, Arun S. Asundi, Ozge D. Bozkurt, Sarah A. Hesse, Christopher J. Takacs, Ravikumar Ramegowda, Jeremy Luterbacher, Chen Eugene, Gregg Beckham, Christopher J. Tassone; “In-Situ Synchrotron-Based Characterization of Soft Materials Properties and Behavior”; September 2023, Bay-area Soft Matter Symposium, Stanford, USA.
22. **Anjani K. Maurya**, Arun S. Asundi, Sarah A. Hesse, Amani M. Ebrahim, Kevin P. Sullivan, Chad T. Palumbo, Allison Z. Werner, Christopher J. Takacs, Gregg T. Beckham, Ritimukta Sarangi, Christopher J. Tassone; “Decoding the mechanism of autoxidation deconstruction reaction of plastics by in-situ simultaneous SAXS and WAXS”; September 2022, SAS2022 International Conference, Campinas, Brazil. (Talk)
21. **Anjani K. Maurya**, René M. Rossi, Alex Dommann, Antonia Neels*; “Unraveling the multiscale structure of fibrous materials by X-ray scattering techniques”; May–June 2022, European Powder Diffraction Conference, EPDIC17, Šibenik, Croatia.
20. **Anjani K. Maurya**, Eloïse Mias, Jean Schoeller, René M. Rossi, Martin Frenz, Alex Dommann, Antonia Neels; “Multiscale structural decoding of electrospun fibers: from processing to possibilities for steering properties”; 30 November 2020; Empa PhD Symposium 2020, Empa, St. Gallen, Switzerland.
19. **Anjani K. Maurya**, Amin Sadeghpour, Ruggero Frison, Riccardo Malini Innocenti, Claudio Toncelli, Marc G. Willinger, René M. Rossi, Martin Frenz, Alex Dommann, Antonia Neels*; “Nucleation and growth study of iron oxide nanoparticles from solution for biomedical application”; Poster presentation, GCB Symposium, January 2020, University of Bern, Switzerland.
18. **Anjani K. Maurya**, Amin Sadeghpour, Ruggero Frison, Riccardo Innocenti Malini, Claudio Toncelli, Marc G. Willinger, René M. Rossi, Martin Frenz, Alex Dommann, Antonia Neels; “Nucleation and growth study of iron oxide nanoparticles from solution”; Poster presentation, Empa PhD Symposium, Switzerland, November 2019.
17. N. D. Tien, **Anjani K. Maurya**, Z. Robert, Giuseppino Fortunato, M. Rottmar, René M. Rossi, Martin Frenz, Alex Dommann, Antonia Neels, Amin Sadeghpour; “Electrospun nanofibers with embedded bioinspired membranes”; Poster presentation, Bioinspired Materials, Monte Verità, October 2019.
16. **Anjani K. Maurya**, Lukas Weidenbacher, Fabrizio Spano, Giuseppino Fortunato, René M. Rossi, Martin Frenz, Alex Dommann, Amin Sadeghpour, Antonia Neels; “Understanding fiber fabrication process by structural feedbacks by SAXS and WAXD”; Poster presentation, European Crystallography Meeting, University of Vienna, 21–22 August 2019.
15. **Anjani K. Maurya**, Lukas Weidenbacher, Fabrizio Spano, Giuseppino Fortunato, René M. Rossi, Martin Frenz, Alex Dommann, Amin Sadeghpour, Antonia Neels; “Decoding structural insights of nanofibers by SAXS and WAXD”; Oral presentation, Swiss Crystallography Meeting, EPFL Sion, Switzerland, August 2019.
14. **Anjani K. Maurya**, A. Sadeghpour, L. Weidenbacher, F. Spano, G. Fortunato, R. M. Rossi, M. Frenz, A. Dommann, A. Neels; “Decoding structural insights into semicrystalline states of polymeric nanofibers at nanoscale”; Poster presentation, Empa PhD Symposium, November 2018.
13. **Anjani K. Maurya**, A. Sadeghpour, L. Weidenbacher, F. Spano, G. Fortunato, R. M. Rossi, M. Frenz, A. Dommann, A. Neels; “Structural-morphological correlation studies in polymeric nanofibers by SAXS and WAXD”; Poster presentation, Polycoll 2018, EPFL, Switzerland, October 2018.
12. **Anjani K. Maurya**, A. Sadeghpour, L. Weidenbacher, F. Spano, G. Fortunato, R. M. Rossi, M.

- Frenz, A. Dommann, A. Neels; “Study the effect of nanofiber fabrication processes on modification of internal structure at nanoscale”; Poster presentation, Swiss Crystallography Meeting 2018, PSI, Switzerland, September 2018.
11. **Anjani K. Maurya**, A. Sadeghpour, L. Weidenbacher, F. Spano, G. Fortunato, R. M. Rossi, M. Frenz, A. Dommann, A. Neels; “Structural insights into semicrystalline states of electrospun nanofibers”; Poster presentation, EXCITE Biomedical Imaging Summer School 2018, ETH Zurich, Switzerland. (Best Poster Award)
 10. **Anjani K. Maurya**, A. Sadeghpour, L. Weidenbacher, F. Spano, G. Fortunato, R. M. Rossi, M. Frenz, A. Dommann, A. Neels; “Structure–morphology correlation studies of electrospun nanofibers by SAXS and WAXS”; Poster presentation, Bio-X, Empa St. Gallen, Switzerland, March 2018.
 9. **Anjani K. Maurya**, A. Sadeghpour, R. Rossi, A. Dommann, A. Neels; “Structural in-situ studies for particle system synthesis and early event dynamics in bio-environments”; Oral presentation, Empa PhD Seminar, Empa St. Gallen, Switzerland, January 2018.
 8. **Anjani K. Maurya**, A. Sadeghpour, R. Rossi, A. Dommann, A. Neels; “Structure–morphology correlation studies in electrospun nanofibers by SAXS and WAXS”; Oral presentation, Empa PhD Symposium 2017, Dübendorf, Switzerland, November 2017.
 7. **Anjani K. Maurya**, A. Sadeghpour, R. Rossi, A. Dommann, A. Neels; “Structural analysis of electrospun fiber membrane by SAXS and WAXS”; Oral presentation, Swiss Crystallography Meeting 2017, Geneva, Switzerland, September 2017.
 6. **Anjani K. Maurya**, N. Saxena, P. Müller-Buschbaum; “In-plane and cross-plane Seebeck coefficients in organic thermoelectric thin films”; 6th Energy Colloquium, Munich School of Engineering, TUM, Germany, July 2016.
 5. **Anjani K. Maurya**, N. Saxena, P. Müller-Buschbaum; “Enhancement of Seebeck coefficient by dedoping of conducting polymer”; Summer School, Austria, June 2016.
 4. **Anjani K. Maurya**, N. Saxena, P. Müller-Buschbaum; “Hybrid Based Thermoelectrics”; MaMa-SELF Status Meeting, oral presentation, Rigi Kulm, Switzerland, May 2016.
 3. **Anjani K. Maurya**, N. Saxena, P. Müller-Buschbaum; “Hybrid Based Thermoelectrics”; Poster presentation, DPG Meeting 2016, Regensburg, Germany, March 2016.
 2. **Anjani K. Maurya**; “Band gap variation in bilayer graphene and silicene under transverse electric field”; Poster presentation, ICANN-2013, IIT Guwahati, India, December 2013.
 1. Lalhriatzuala, **Anjani K. Maurya**, Pratima Agarwal; “Structural and transport studies of ZnS thin film prepared by Doctor Blade’s technique”; Poster presentation, Condensed Matter Days Conference 2011, Guwahati University, India, August 2011.